

# The doubly interval-valued Birkhoff inequalities: endpoint reduction and the reverse-Hölder obstruction

Candidate full solution to the conclusion question in arXiv:2507.16332

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**Result.** When both the integrand  $F = [f_-, f_+]$  and set function  $M = [\nu_-, \nu_+]$  are interval-valued, their Birkhoff weak integral factors exactly into the two scalar endpoint integrals. Consequently Hölder ( $p, q > 1$ ), Minkowski ( $p \geq 1$ ), and reverse Minkowski ( $0 < p < 1$ ) hold componentwise in the weak interval order. For reverse Hölder, the natural interval  $q$ -gauge fails even on one atom: at  $p = 1/2$  the desired inequality would require  $[1, 4] \succeq [1/4, 16]$ . There is nevertheless an unconditional sharp diagonal-hull form. If  $\nu_+ \leq C\nu_-$ , the natural form is restored with the optimal factor  $C^{1/q}$ , and no positive width-independent factor exists.

## 1 The source question

Croitoru, Iosif, Sambucini, and Zampogni [1] prove scalar Birkhoff weak Hölder and Minkowski inequalities for non-additive set functions, along with their reverses. They extend the results separately to interval-valued integrands and to interval-valued set functions. Their conclusion asks for the remaining case where *both* objects are interval-valued:

### 5. CONCLUSION

We have proved several inequalities and reverse forms of the classical Hölder and Minkowski inequalities for Birkhoff weakly integrable functions when the underlying set function with respect to which we integrate is non-additive. Moreover, these results have also been extended to vector-valued integrands and to the multivalued setting, where the integrands, or the set functions involved, are allowed to take interval-valued outputs.

Hölder's and Minkowski's inequalities remains an open problem when both the function  $F$  and the set functions  $M$  are interval-valued. In this latter setting, results concerning integrability, convergence, and properties of the integral have been established, while the corresponding inequalities are currently the subject of ongoing investigation. Also applications in Image Reconstruction, in the spirit of Remark 4.11 and of Section 4.3, will be exploited in a future study.

Figure 1: The open problem in the conclusion of arXiv:2507.16332.

The same authors had previously treated an analogous doubly interval-valued Riemann–Lebesgue setting [2]. The question here is specifically for the Birkhoff weak integral and the scalar theorems proved in [1].

## 2 Setup and proof intuition

Let  $\mathcal{I}_+$  denote the nonempty compact intervals in  $\mathbb{R}_+$ . Use Minkowski addition and multiplication,

$$[a, b] \oplus [c, d] = [a + c, b + d], \quad [a, b] \otimes [c, d] = [ac, bd],$$

and the weak order  $[a, b] \preceq [c, d]$  iff  $a \leq c$  and  $b \leq d$ . Let

$$M(A) = [\nu_-(A), \nu_+(A)], \quad \nu_-(A) \leq \nu_+(A), \tag{1}$$

where each endpoint belongs to the source class  $\mathcal{M}_{cs}(\mathcal{E})$ : it is nonnegative, subadditive, continuous from below, and  $\mathcal{E}$ -integrable in the source's sense.

For  $F(t) = [f_-(t), f_+(t)] \in \mathcal{I}_+$ , define its doubly interval-valued Birkhoff weak integral by convergence in Hausdorff distance of the tagged sums

$$\bigoplus_{k=1}^n F(t_k) \otimes M(B_k). \quad (2)$$

The proof idea is visible before taking a limit: every sum in (2) is exactly

$$\left[ \sum_{k=1}^n f_-(t_k) \nu_-(B_k), \sum_{k=1}^n f_+(t_k) \nu_+(B_k) \right]. \quad (3)$$

Hausdorff convergence of intervals is coordinatewise convergence. Thus the whole problem splits into the two scalar channels  $(f_-, \nu_-)$  and  $(f_+, \nu_+)$ . Positive powers preserve endpoints, so the standard Hölder and both Minkowski regimes follow immediately. Negative powers reverse endpoints; that single reversal is exactly why natural reverse Hölder fails.

### 3 Exact endpoint factorization

**Theorem 1** (Endpoint theorem). *The integral in (2) exists if and only if  $f_- \in B_w(\nu_-, T)$  and  $f_+ \in B_w(\nu_+, T)$ . In that case*

$$\int_T F dM = \left[ \int_T f_- d\nu_-, \int_T f_+ d\nu_+ \right]. \quad (4)$$

*The same assertion holds for every endpoint expression below.*

*Proof.* Identity (3) follows from interval arithmetic. If the interval sums converge to  $[a, b]$  in Hausdorff distance, their lower and upper endpoints converge uniformly over all sufficiently fine tagged partitions to  $a$  and  $b$ , which is precisely scalar Birkhoff weak integrability. Conversely, choose the common refinement of scalar control partitions for the two endpoints and the maximum of their tail indices. Both endpoint errors are then below  $\varepsilon$ , so their maximum, the Hausdorff error, is below  $\varepsilon$ . Passing to the limits in (3) gives (4).  $\square$

For  $r > 0$  and all required endpoint powers integrable, put

$$\mathfrak{N}_{r,M}(F) := \left[ \left( \int f_-^r d\nu_- \right)^{1/r}, \left( \int f_+^r d\nu_+ \right)^{1/r} \right]. \quad (5)$$

This is an interval: the corresponding tagged sums for the lower channel are bounded by those of the upper channel.

**Theorem 2** (Complete positive-power theory). *Let  $F = [f_-, f_+]$  and  $G = [g_-, g_+]$  be nonnegative and assume every scalar endpoint expression displayed is Birkhoff weak integrable.*

1. *If  $p, q > 1$  and  $p^{-1} + q^{-1} = 1$ , then*

$$\mathfrak{N}_{1,M}(F \otimes G) \preceq \mathfrak{N}_{p,M}(F) \otimes \mathfrak{N}_{q,M}(G). \quad (6)$$

2. *If  $p \geq 1$ , then*

$$\mathfrak{N}_{p,M}(F \oplus G) \preceq \mathfrak{N}_{p,M}(F) \oplus \mathfrak{N}_{p,M}(G). \quad (7)$$

3. *If  $0 < p < 1$ , then the direction reverses:*

$$\mathfrak{N}_{p,M}(F \oplus G) \succeq \mathfrak{N}_{p,M}(F) \oplus \mathfrak{N}_{p,M}(G). \quad (8)$$

*Here the integrability assumptions are exactly the scalar hypotheses from [1], applied once to each endpoint channel.*

*Proof.* Theorem 1 turns each side into an interval whose lower endpoint is the scalar expression for  $(f_-, g_-, \nu_-)$  and whose upper endpoint is the corresponding expression for  $(f_+, g_+, \nu_+)$ . Apply the source's scalar Hölder, Minkowski, or reverse Minkowski theorem in the two channels. The two resulting inequalities are exactly the weak interval orders (6)–(8).  $\square$

## 4 Reverse Hölder: failure, sharp repair, and width factor

Fix  $0 < p < 1$  and  $q = p/(p-1) < 0$ , and suppose  $g_- > 0$ . Natural interval functional calculus gives

$$G^q = [g_+^q, g_-^q], \quad \mathfrak{N}_{q,M}^{\text{nat}}(G) := \left( \int G^q dM \right)^{1/q} = [\|g_-\|_{q,\nu_+}, \|g_+\|_{q,\nu_-}], \quad (9)$$

where  $\|h\|_{r,\nu} = (\int h^r d\nu)^{1/r}$ . The crossed measures in (9) are forced by the two order reversals.

**Proposition 1** (The natural reverse fails maximally). *In general,*

$$\mathfrak{N}_{1,M}(F \otimes G) \succeq \mathfrak{N}_{p,M}(F) \otimes \mathfrak{N}_{q,M}^{\text{nat}}(G) \quad (10)$$

*is false, even for additive endpoint measures concentrated at one atom and constant singleton intervals  $F = G = [1, 1]$ . No positive universal constant can be inserted before the right side of (10).*

*Proof.* On  $T = \mathbb{N}$ , concentrate  $\nu_-$  and  $\nu_+$  at 1 with masses  $a$  and  $b$ , where  $0 < a < b$ . Constant integrands make every tagged sum exact. The two sides of (10) are

$$[a, b], \quad [a^{1/p}b^{1/q}, b^{1/p}a^{1/q}]. \quad (11)$$

For  $p = 1/2, q = -1, a = 1, b = 4$ , these are  $[1, 4]$  and  $[1/4, 16]$ , so the upper endpoint inequality fails. More generally, the ratio of the left upper endpoint to the right upper endpoint is  $(b/a)^{1/q} \rightarrow 0$  as  $b/a \rightarrow \infty$ . This excludes every positive width-independent factor.  $\square$

The failure has a sharp unconditional repair. For two numbers let  $\text{Hull}(x, y) = [\min(x, y), \max(x, y)]$ , and define

$$\mathcal{D}_{p,q}(F, G; M) = \text{Hull}(\|f_-\|_{p,\nu_-} \|g_-\|_{q,\nu_-}, \|f_+\|_{p,\nu_+} \|g_+\|_{q,\nu_+}). \quad (12)$$

**Theorem 3** (Sharp reverse Hölder replacements). *Under the scalar reverse-Hölder integrability hypotheses,*

$$\boxed{\mathfrak{N}_{1,M}(F \otimes G) \succeq \mathcal{D}_{p,q}(F, G; M)}. \quad (13)$$

*This is exact on one-atom constant examples. Moreover, if for some  $C \geq 1$*

$$\nu_- \leq \nu_+ \leq C\nu_- \quad (14)$$

*pointwise on  $\mathcal{E}$ , then the natural gauge satisfies*

$$\boxed{\mathfrak{N}_{1,M}(F \otimes G) \succeq C^{1/q}(\mathfrak{N}_{p,M}(F) \otimes \mathfrak{N}_{q,M}^{\text{nat}}(G))}. \quad (15)$$

*The factor  $C^{1/q}$  is optimal.*

*Proof.* Scalar reverse Hölder in the two matching channels gives

$$L := \int f_- g_- d\nu_- \geq R_-, \quad U := \int f_+ g_+ d\nu_+ \geq R_+,$$

where  $R_-$  and  $R_+$  are the two numbers in (12). Since  $L \leq U$ , we have  $L \geq \min(R_-, R_+)$  and  $U \geq \max(R_-, R_+)$ , proving (13).

For (15), positive tagged sums and (14) imply  $\int h d\nu_+ \leq C \int h d\nu_-$  for every relevant  $h \geq 0$ . As  $1/q < 0$ ,

$$\|g_+\|_{q,\nu_+} \geq C^{1/q} \|g_+\|_{q,\nu_-}, \quad \|g_-\|_{q,\nu_-} \geq \|g_-\|_{q,\nu_+}.$$

Insert the first estimate into the upper-channel scalar reverse Hölder inequality. The second estimate proves the lower endpoint inequality even with factor 1, hence also with  $C^{1/q} \leq 1$ . This proves (15). In (11), take  $b = Ca$ ; the upper endpoints in (15) are equal, proving optimality.  $\square$

## 5 Verification, novelty, and scope

The included script checked 2,000 random finite-atomic double-interval models for (6), both Minkowski directions, and the diagonal-hull reverse Hölder inequality. It also checks the exact  $[1, 4]$  versus  $[1/4, 16]$  obstruction and equality for the factor  $C^{1/q} = 1/4$ . These are sanity checks; all proofs above are endpoint identities plus the source’s scalar theorems.

A bounded novelty search on 17 August 2026 covered the run’s cheap indexes, the exact source title and question, and double interval-valued Birkhoff, Hölder, and Minkowski combinations. The closest prior result is the Riemann–Lebesgue theory in [2], which treats the analogous endpoint calculus for that different integral. The 2026 published version of [1] still states the Birkhoff weak problem as open. No later direct answer was found. Novelty is plausible, not certified.

Equations (6) and (7) fully answer the named positive-exponent question. Equations (8), (13), and (15), together with Proposition 1, give the complete natural endpoint picture for the reverse regime. Human review should audit that the double Birkhoff definition in (2) matches the intended established definition and that the endpoint set functions satisfy the same  $\mathcal{M}_{cs}$  hypotheses as the source scalar theorems.

## References

- [1] A. Croitoru, A. Iosif, A. R. Sambucini, and L. Zampogni, *Some Reverse Inequalities for Scalar Birkhoff Weak Integrable Functions*, arXiv:2507.16332v2; *Axioms* **15** (2026), 133.
- [2] A. Croitoru, A. Gavriluț, A. Iosif, and A. R. Sambucini, *Inequalities in Riemann–Lebesgue Integrability*, *Mathematics* **12** (2024), 49.
- [3] E. Pap, A. Iosif, and A. Gavriluț, *Integrability of an Interval-Valued Multifunction with Respect to an Interval-Valued Set Multifunction*, *Iranian J. Fuzzy Systems* **15** (2018), 47–63.